MA 232 Course Project 2:
How AI Detects Objects: A Linear Algebra Approach to Pattern Recognition

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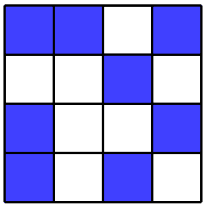
"I pledge my honor that I have abided by the Stevens Honor System."

Part A: Convert the Pattern into a Vector

Task A.1: Target Grid and Vector Representation

Convert the given grid into a vector $w \in \mathbb{R}^{16}$.

Answer: The target pattern is a 4×4 binary grid with blue cells as 1 and white cells as 0:



Reading row by row from top to bottom (left to right) yields the target vector $w \in \mathbb{R}^{16}$:

$$w = \begin{bmatrix} 1 & 1 & 0 & 1 & 0 & 0 & 1 & 0 & 1 & 0 & 0 & 1 & 1 & 0 & 1 & 0 \end{bmatrix}^T$$

Task A.2: Cell Counts

How many blue cells and white cells does the target pattern contain?

Answer: The target pattern contains **8 blue cells** (value 1) and **8 white cells** (value 0), for a total of 16 cells.

Task A.3: General Pattern Encoding Procedure

Explain how an arbitrary blue-and-white 4×4 grid can be represented by a vector $u \in \{0, 1\}^{16}$.

Answer: We can treat the 4×4 grid like a coordinate system where each cell has a row i and column j (from 1 to 4). Each cell takes a value of 1 if it is filled with blue, or 0 if it is white.

To flatten this into a vector, we read the grid row by row from top to bottom (left to right) into a 16-element column vector $u \in \{0,1\}^{16}$. Because there are 16 cells and each cell can be either 0 or 1, this vector representation accounts for all $2^{16} = 65,536$ possible blue-and-white patterns.

Part B: Construct a Matrix for Pattern Recognition

Task B.1: Information Represented by Vector u

Explain what information is represented by the vector u.

Answer: The vector $u \in \{0,1\}^{16}$ represents any 4×4 test pattern we want to check against our target pattern. Each of the 16 entries in u tells us whether that specific box on the grid is blue (1) or white (0).

Task B.2: Role of Quadratic Form $Q(u) = u^T Mu$

Describe the role of the quadratic form $Q(u) = u^T Mu$ in recognizing a pattern.

Answer: The quadratic form $Q(u) = u^T Mu$ works like a scoring function for any test pattern u:

- If the function returns 0, the test pattern matches our target pattern ($Q(w) = w^T Mw = 0$).
- If the function returns a nonzero (positive) score, the pattern is not the target pattern ($Q(u) \neq 0$).

This turns pattern recognition into a simple check: an input pattern is recognized as the target pattern if and only if $Q(u) = 0$.

Task B.3: Construction of Matrix M

Construct a 16×16 matrix M that satisfies the two conditions: $w^T Mw = 0$ and $u^T Mu \neq 0$ for every other nonzero binary vector $u \neq w$.

Answer: We construct M in the form $M = B^T B$. To guarantee that $Mw = \mathbf{0}$, we choose the rows of B to be 15 vectors $v_1, \ldots, v_{15}$ that are all perpendicular (orthogonal) to w.

1. Setting the dot product $w \cdot v = 0$ using our target vector gives:

$$v_1 + v_2 + v_4 + v_7 + v_9 + v_{12} + v_{13} + v_{15} = 0$$

2. Solving for the first component:

$$v_1 = -(v_2 + v_4 + v_7 + v_9 + v_{12} + v_{13} + v_{15})$$

   Setting each of the other 15 free components to 1 (one at a time, with all others 0) produces a basis of 15 vectors $\{v_1, v_2, \ldots, v_{15}\}$ that are all orthogonal to w.

3. We stack these 15 vectors as rows of a 15×16 matrix B:

$$B = \begin{bmatrix} v_1^T \\ v_2^T \\ \vdots \\ v_{15}^T \end{bmatrix}$$

4. We set $M = B^T B$, which produces our 16×16 matrix.

Task B.4: Verification of the Null Target Condition

Show that your matrix satisfies $w^T M w = 0$.

Answer: Because every row of B was chosen to be orthogonal to w, multiplying B by w gives:

$$Bw = \begin{bmatrix} v_1^T w \\ \vdots \\ v_{15}^T w \end{bmatrix} = \begin{bmatrix} 0 \\ \vdots \\ 0 \end{bmatrix} = \mathbf{0}$$

This means $Mw = B^T(Bw) = B^T\mathbf{0} = \mathbf{0}$.

Evaluating the quadratic form for the target vector gives:

$$w^T M w = w^T(Mw) = w^T\mathbf{0} = 0$$

Task B.5: Nonzero Condition for Non-Target Binary Vectors

Explain why the quadratic form must be nonzero for every other nonzero binary vector.

Answer: Using the transpose product rule, the quadratic form can be written as:

$$u^T M u = u^T(B^T B)u = (Bu)^T(Bu) = \|Bu\|^2$$

Because $\|Bu\|^2$ is a squared length, it is always ≥ 0, and equals 0 if and only if $Bu = \mathbf{0}$. The only vectors satisfying $Bu = \mathbf{0}$ are scalar multiples of the target pattern, $u = c \cdot w$.

Since u is restricted to binary patterns (0 and 1 only):

- $c = 0 \implies u = \mathbf{0}$ (the all-white grid).
- $c = 1 \implies u = w$ (the target pattern).

For any other nonzero binary vector $u \neq w$, u is not a multiple of w, which means $Bu \neq \mathbf{0}$. This guarantees that the score is strictly positive:

$$u^T M u = \|Bu\|^2 > 0 \neq 0$$

Task B.6: Symmetry and Invertibility of M

State whether your matrix is symmetric and whether it is invertible. Justify your answers.

Answer:

- **Symmetry: Yes.** Using the transpose product rule:

$$M^T = (B^T B)^T = B^T (B^T)^T = B^T B = M$$

  Since $M^T = M$, matrix M is symmetric.

- **Invertibility: No.** From Task B.4, we know that $Mw = \mathbf{0}$. Because matrix M maps a nonzero vector ($w \neq \mathbf{0}$) to the zero vector, w is a non-trivial vector in the nullspace of M. Therefore, $\det(M) = 0$, its rank is 15 (not full rank 16), and M is not invertible.

Part C: Verify the Recognition Rule

Task C.1: Systematic Verification Methodology

Develop a systematic method to verify your matrix.

Task C.2: Verification of the Target Vector

Verify the quadratic-form value for the target vector.

Task C.3: Testing on Required Binary Patterns

Test your matrix on several other binary patterns, including: a grid with only one blue cell, a grid with two blue cells, a grid with three blue cells in different positions, and the grid with all 16 cells blue.

Task C.4: Exhaustive Verification of All $2^{16} - 1 = 65,535$ Nonzero Binary Vectors

Explain how you can verify the condition for all $2^{16} - 1 = 65,535$ nonzero binary vectors.

Task C.5: Table of Test Patterns and Values

Present a table of at least 10 test patterns, their binary vectors, and the corresponding quadratic-form values.

Table 1 presents the quadratic-form evaluation values generated by `calculate.py` for both Matrix M (constructed via basis of $W^{\perp}$) and Matrix N (constructed via orthogonal projection).

#	Description	Blue Cells	$Q_M(u) = u^T M u$	$Q_N(u) = u^T N u$
1	Target pattern w	8	0	0
2	Single blue cell (Cell 1, in w)	1	7	7
3	Single blue cell (Cell 3, not in w)	1	1	8
4	Two blue cells (Cells 1, 2, both in w)	2	6	12
5	Two blue cells (Cells 1 in w, 3 not in w)	2	8	15
6	Three blue cells (Row 1: 1, 2, 4 in w)	3	5	15
7	Three blue cells (Cells 3, 5, 6, none in w)	3	3	24
8	Four target cells (Cells 1, 2, 4, 7)	4	4	16
9	Inverted target $(1 - w)$	8	8	64
10	All 16 cells blue	16	8	64

Table 1: Test patterns and quadratic-form evaluation values.

Task C.6: Generality Beyond the Sample Table

State how you know that your construction satisfies the requirement for all nonzero binary patterns, not merely the patterns in your sample table.

Part D: Explore Different Matrix Constructions

D.1: Construct a Second Matrix

Task D.1.a: Construction of Second Matrix N

Find a second 16×16 *matrix* N*, different from your original matrix* M*, that recognizes the same target pattern.*

Task D.1.b: Verification of Conditions for Matrix N

Verify that $w^T N w = 0$ *and* $u^T N u \neq 0$ *for every other nonzero binary vector* $u \neq w$*.*

Task D.1.c: Comparison Between Matrices M and N

Explain how the second matrix differs from the first and why both matrices work.

D.2: Investigate Matrix Structure

Task D.2.a: Viability of Off-Diagonal Entries

Can a matrix with nonzero off-diagonal entries satisfy the requirements?

Task D.2.b: Contribution of Matrix Entries to $u^T M u$

Explain how the entries of a matrix contribute to the value of $u^T M u$*.*

Task D.2.c: Construction/Description of Non-Diagonal Matrix

Construct or describe a non-diagonal matrix that satisfies the recognition condition, if possible.

Part E: Reflection and Applications

(a) Converting a Visual Grid into a Vector

How does converting a visual grid into a vector make mathematical analysis possible?

(b) Challenges in Constructing the Matrix

What was the most challenging part of constructing the matrix?

(c) Verifying Pattern Uniqueness

How did you verify that the matrix recognizes only the target pattern?

(d) Importance of Systematic Verification

Why is systematic verification important in pattern recognition?

(e) Real-World Applications

How could a similar approach be used in image processing, data science, or another real-world application?